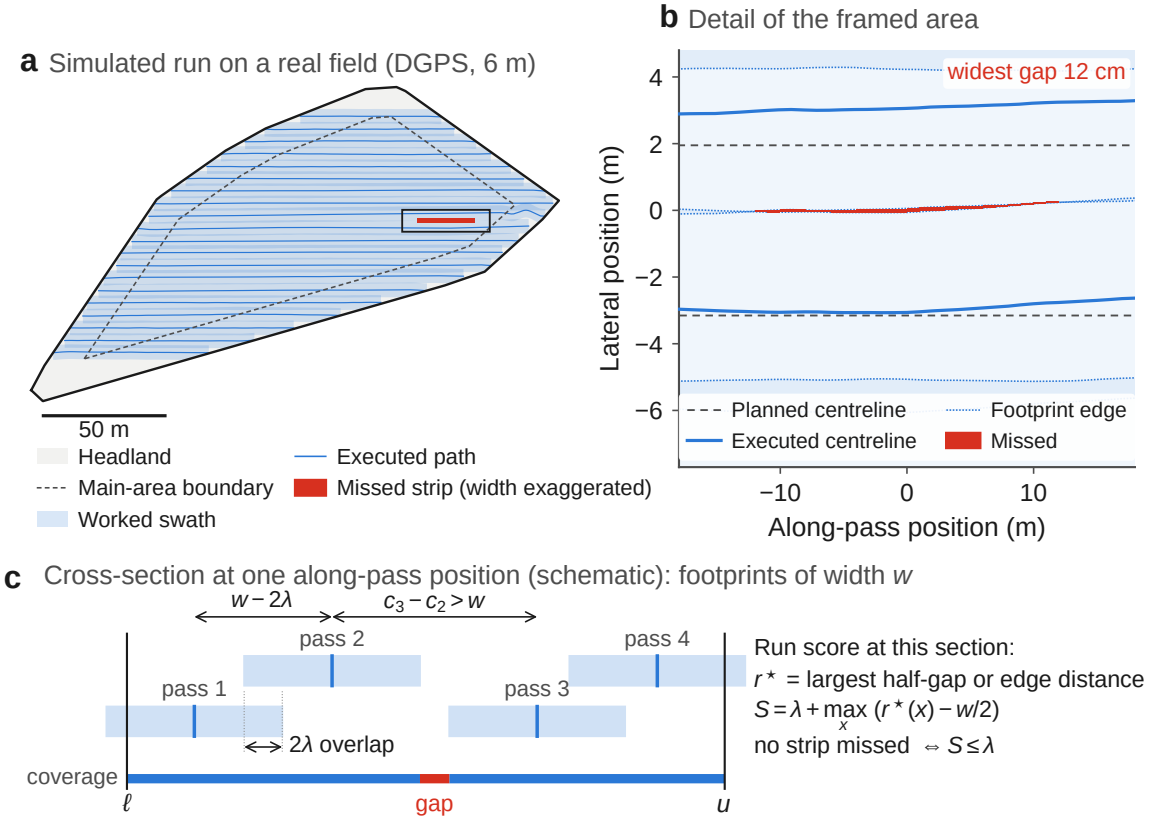

a Simulated run on a real field (DGPS, 6 m)
50 m
Headland
Main-area boundary
Worked swath
Executed path
Missed strip (width exaggerated)
b Detail of the framed area
widest gap 12 cm
Lateral position (m)
4
2
0
−2
−4
−6
−10
0
10
Along-pass position (m)
Planned centreline
Executed centreline
Footprint edge
Missed
c Cross-section at one along-pass position (schematic): footprints of width w
w − 2λ
c3 − c2 > w
pass 1
pass 2
pass 3
pass 4
2λ overlap
coverage
ℓ
gap
u
Run score at this section:
r⋆ = largest half-gap or edge distance
S = λ + max_x (r⋆(x) − w/2)
no strip missed ⇔ S ≤ λ